TSAR Project Part 1 – Exploratory Data Analysis

Kryštof Vávra, Said Fakhri (Group 03)

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```r
# # Random sample your group's data set from the original LCdata data set
# LCdata <- fread("LCdata.csv")
# set.seed(1)
# myLCdata <- LCdata %>% sample(5) %>% slice_sample(prop = .5)
# fwrite(myLCdata, file = "myLCdata.csv")
myLCdata <- fread("myLCdata.csv")
```

1 Task 1: Business Understanding

```r
glimpse(myLCdata)
```

```
Rows: 44,369
Columns: 5
$ annual_inc     <dbl> 60000, 18576, 40000, 35000, 125000, 144000, 205000, 118~
$ open_acc       <int> 6, 7, 15, 3, 12, 8, 9, 18, 12, 3, 11, 10, 15, 5, 13, 10~
$ purpose        <chr> "credit_card", "debt_consolidation", "small_business", ~
$ home_ownership <chr> "OWN", "RENT", "RENT", "OWN", "RENT", "MORTGAGE", "MORT~
$ inq_last_6mths <int> 0, 0, 2, 1, 0, 2, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 3, 0~
```

```r
str(myLCdata)
```

```
Classes 'data.table' and 'data.frame':  44369 obs. of  5 variables:
 $ annual_inc    : num  60000 18576 40000 35000 125000 ...
 $ open_acc      : int  6 7 15 3 12 8 9 18 12 3 ...
 $ purpose       : chr  "credit_card" "debt_consolidation" "small_business" "home_improvemen
```

```
 $ home_ownership: chr  "OWN" "RENT" "RENT" "OWN" ...
 $ inq_last_6mths: int  0 0 2 1 0 2 0 0 1 0 ...
 - attr(*, ".internal.selfref")=<externalptr>
```

1.1 Variable Overview and Business Interpretation

The dataset `myLCdata` contains 44,369 observations and 5 variables. Based on the output of `glimpse()` and `str()`, the variables can be categorized as follows:

1.1.1 Variable Description

- **annual_inc (numeric)**
  The self-reported annual income provided by the borrower during registration. This variable is continuous and measures repayment capacity.
- **open_acc (integer)**
  The number of open credit lines in the borrower's credit file. This reflects the borrower's existing credit exposure.
- **purpose (character)**
  A category provided by the borrower for the loan request. (e.g., credit card refinancing, debt consolidation, small business, home improvement). This is a categorical variable.
- **home_ownership (character)**
  The home ownership status provided by the borrower during registration.(e.g., RENT, OWN, MORTGAGE, OTHER).
- **inq_last_6mths (integer)**
  The number of inquiries in past 6 months (excluding auto and mortgage inquiries). This is an indicator of recent credit-seeking behavior and potential financial stress.

1.2 Business Context in Peer-to-Peer Lending

<what to write here??????????>

2 Task 2: Data Understanding Based on Summary Statistics

2.1 Summary Statistics

```r
summary(myLCdata)
```

```
   annual_inc         open_acc       purpose           home_ownership
 Min.   :   4000   Min.   : 1.00   Length:44369       Length:44369
 1st Qu.:  45649   1st Qu.: 8.00   Class :character   Class :character
 Median :  65000   Median :11.00   Mode  :character   Mode  :character
 Mean   :  75269   Mean   :11.57
 3rd Qu.:  90000   3rd Qu.:14.00
 Max.   :6000000   Max.   :76.00
                   NA's   :2
 inq_last_6mths
 Min.   : 0.0000
 1st Qu.: 0.0000
 Median : 0.0000
 Mean   : 0.6966
 3rd Qu.: 1.0000
 Max.   :31.0000
 NA's   :2
```

```r
# install skimr if necessary
if (!require("skimr")) install.packages("skimr")
library(skimr)

skim(myLCdata)
```

Table 1: Data summary

Name	myLCdata
Number of rows	44369
Number of columns	5
Key	NULL
Column type frequency:	
character	2
numeric	3

Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
purpose	0	1	3	18	0	14	0
home_ownership	0	1	3	8	0	6	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
annual_inc	0	1	75269.01	60410.50	4000	45649	65000	90000	6000000	
open_acc	2	1	11.57	5.38	1	8	11	14	76	
inq_last_6mths	2	1	0.70	1.01	0	0	0	1	31	

2.2 Interpretation and Domain Context

2.2.1 1. Missing Values (NAs) ——

- Are missing values present?
  - Categorical values: No missing values
  - Numerical values:
    - Missing values: 2 tables open_acc, 2 inq_last_6mths
    - No missing values: annual_inc
- Is this plausible from a business perspective?
- Would this require data preparation?
  - Yes, but can be handled by removing the data objects. The risk of distorting of the data is low due to small number of data objects.

Discussion:

With such a low number missing values and the total number of data objects of 44369 it the easiest way to handle the missing values is to remove the data objects with missing values. imputation methods require pre calculations which in this case are not worth to do.

2.2.2 2. Numerical Variables

- Mean / Median:
  - The difference in mean & median seems high in the table inq_last_6mths given the distribution of the 1st Qu and 3rd Qu. This can be explained by the Max which 31 and also an outlier.
- Minimum:
  - The minimum seems reasonable in all the tables
- Maximum:
  - The maximum in annual_inc and inq_last_6mths seem to be outlier. Maybe these data objects can be removed to reduce the distortion in this tables.
- Standard deviation:
  - The std. dev. in the annual_inc table seems too high with a value of 60410. The reason could lay in the outlier of the max which is 6 mio.
- Is the distribution symmetric or skewed?
  - symmetric:
    - open_acc
  - right skewed:
    - annual_inc, inq_last_6mths
- Are there potential outliers?
  - The min of 4000 and max of 6 mio seems to be outliers in the table annual_inc
  - The max of 31 in inq_last_6mths seems to be also an outlier
- Do the extreme values seem realistic in the Lending Club context?
  - The outlier of 6 mio in the table annual_inc does not the be realistic in the lending club context. You should expect a person having an income of 6 mio does need a loan from a platform. The other outliers seem to be realistic in this context.

Statistical Interpretation: > Write here.

Domain Interpretation: > Write here.

2.2.3 3. Categorical Variables

For each categorical variable:

- Number of categories:
  - purpose: 14
  - home_ownership: 6
- Most frequent category:
  - is not readable from the summary or skim table
- Least frequent category:
  - is not readable from the summary or skim table
- Are categories balanced or highly unbalanced?
  - is not readable from the summary or skim table

Statistical Interpretation: > Write here.

Domain Interpretation: > Write here.

2.2.4 4. Key Insights from Summary Statistics

- Most important statistical findings:
- Potential data quality issues:
- First hypotheses about borrower behavior:

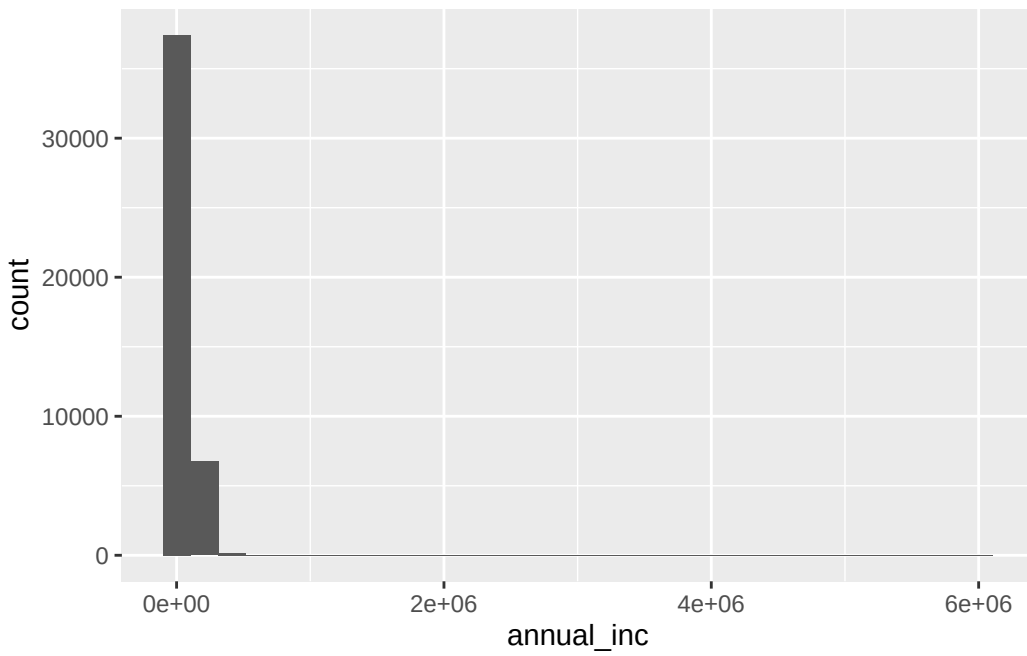
---

3 Task 3: Univariate Exploratory Data Analysis (EDA) with ggplot2

3.1 Variable 1: annual_inc

3.1.1 Visualization

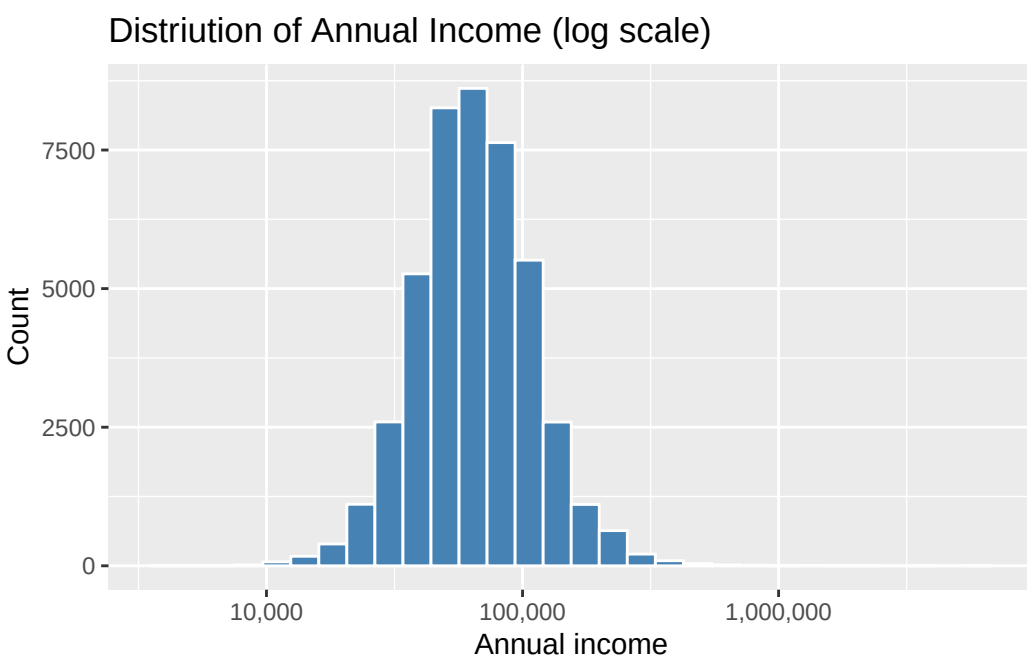
```
ggplot(myLCdata, aes(x = annual_inc)) +
  geom_histogram(bins = 30)
```



3.1.2 Interpretation:

To visualize the annual_inc feature, a histogram was chose because it is suitable for dislaying the distribution of continuous numerical variables The distribution of annual income is extremly right-skewed. Most observations lie in the lower income range, while a small number of observations have extremely high income values (e.g., up to approximately 6 million). These extreme values were identified during Task 2. Such outliers stretch the scale of the histogram and make it difficult to interpret the distribution of the majority of observations. Therefore, a logarithmic scale was applied to the x-axis. The log transformation compresses large values and spreads smaller values, allowing the overall distribution of incomes to be visualized more clearly.:

```r
ggplot(myLCdata, aes(x = annual_inc)) +
  geom_histogram(bins = 30, color = "white", fill = "steelblue") +
  scale_x_log10(labels = label_comma()) +
  labs(
    title = "Distriution of Annual Income (log scale)",
    x = "Annual income",
    y = "Count"
  )
```

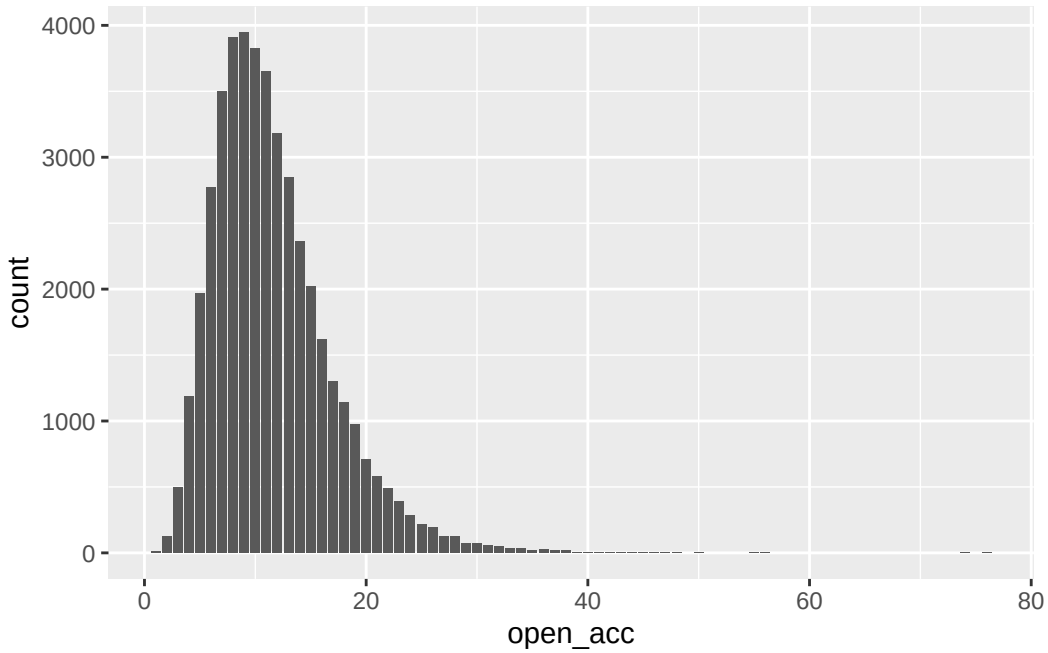


3.2 Variable 2: `open_acc`

3.2.1 Visualization

```r
ggplot(myLCdata, aes(x = open_acc))+
  geom_bar()
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_count()`).
```

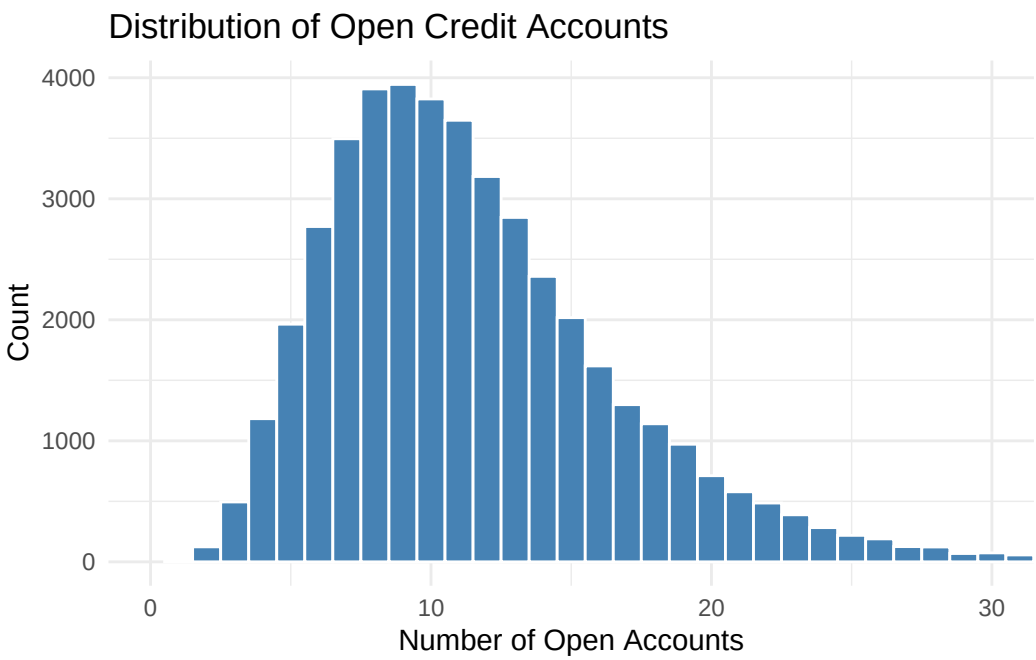


3.2.2 Interpretation

To visualize the open_acc variable, a histogram was used because it represents a numerical count variable. The distribution is right-skewed, meaning that most borrowers have a relatively small number of open credit accounts, while a smaller number of borrowers have a significantly higher number of accounts. As identified in Task 2, the highest value represents an outlier. Most borrowers have between approximately 5 and 15 open accounts, which represents the main concentration of the data. To better illustrate this distribution and improve readability, the histogram focuses on the relevant range of the data (0-30) to cancel the influence of extreme values.:

```
ggplot(myLCdata, aes(x = open_acc)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  coord_cartesian(xlim = c(0, 30)) +
  labs(
    title = "Distribution of Open Credit Accounts",
    x = "Number of Open Accounts",
    y = "Count"
  ) +
  theme_minimal()
```

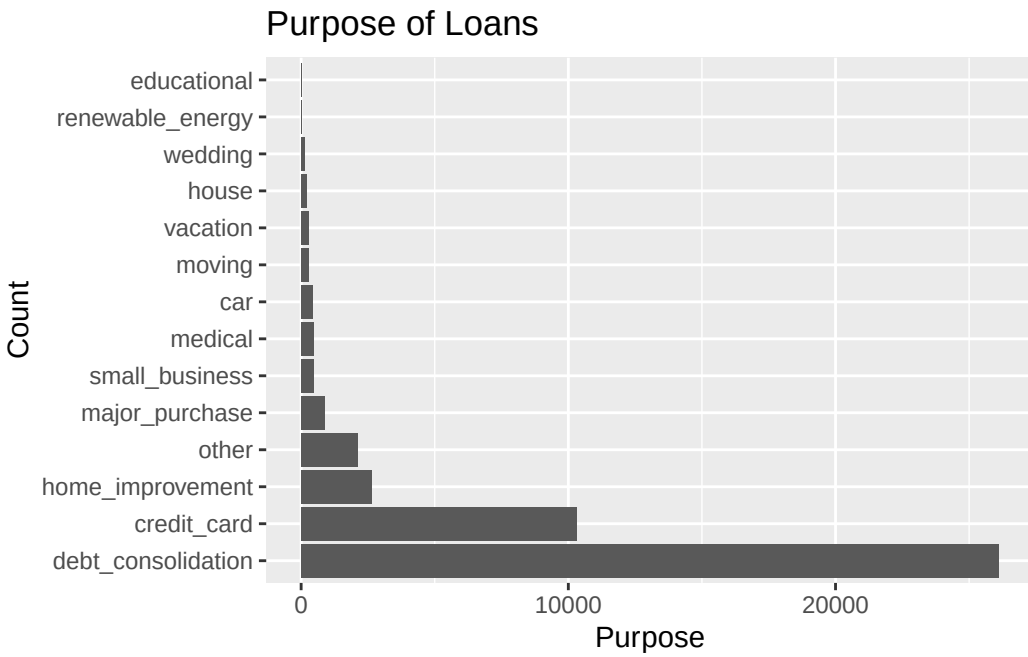
```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).
```



3.3 Variable 3: `purpose`

3.3.1 Visualization

```
ggplot(myLCdata, aes(x = forcats::fct_infreq(purpose))) +
  geom_bar() +
  coord_flip() +
  labs(
    title = "Purpose of Loans",
    x = "Count",
    y = "Purpose"
  )
```



3.3.2 Interpretation

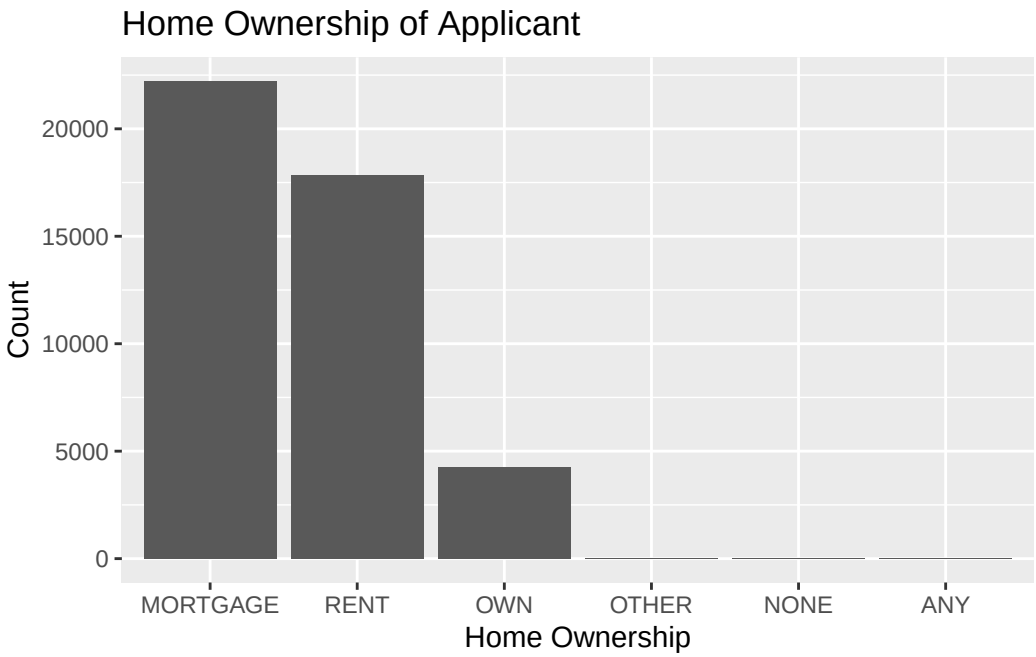
To visualize the purpose variable, a bar chart was used. The chart shows the number of loans for each loan purpose. The visualization indicates that debt consolidation is by far the most common reason for taking a loan on the platform. The second most common purpose is credit card repayment, followed by home improvement and other smaller categories. According to the task description, the platform is a peer-to-peer lending marketplace that may offer more competitive loan conditions compared to traditional lenders. This may explain why many borrowers use the platform to consolidate existing debts into a single loan with potentially better terms. Credit Cards supplier are also known for their extrem interest.

3.4 Variable 4: `home_ownership`

3.4.1 Visualization

```
ggplot(myLCdata, aes(x = forcats::fct_infreq(home_ownership)))+
  geom_bar()+
  labs(
    title = "Home Ownership of Applicant",
```

```
    x = "Home Ownership",
    y = "Count"
  )
```

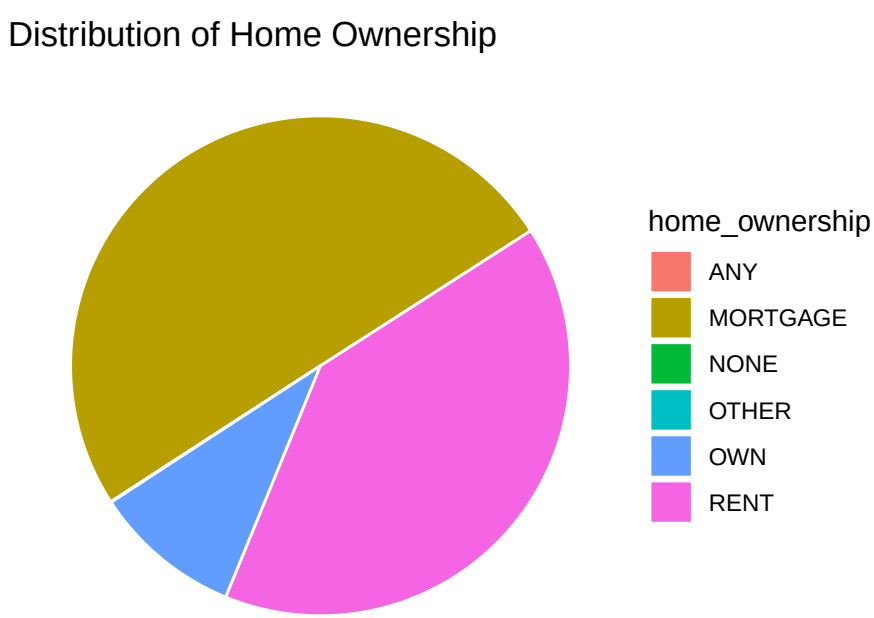


3.4.2 Interpretation

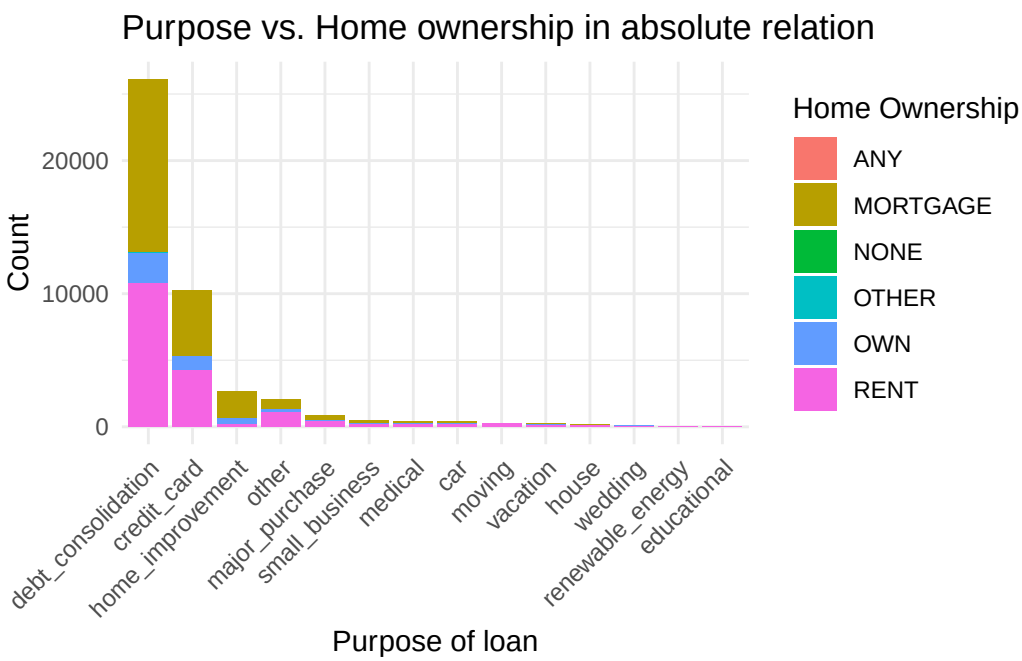
To visualize the home_ownership variable, a pie chart was used because it shows the proportion of observations within a categorical variable. The chart shows that the most common categories are mortgage and rent, while a smaller proportion of borrowers own their homes outright. The remaining categories (such as ANY, NONE, and OTHER) appear only rarely and therefore represent a small share of the dataset. One possible explanation for the relatively small proportion of borrowers who fully own their homes is that homeowners may have easier access to traditional bank loans, as property can serve as collateral. As a result, they may rely less on peer-to-peer lending platforms compared to individuals who rent or still have a mortgage. Interesting to note is, that on the feature "purpose" home improvements were at third place. Here ownership of home is also on the third place. This may indicate that some homeowners use loans on the platform to finance improvements or renovations.

```r
ggplot(myLCdata, aes(x="",fill = home_ownership))+
  geom_bar(width = 1, color = "white")+
  coord_polar("y", start=1)+
  labs(
    title = "Distribution of Home Ownership",
    file = "Home Ownership"
  )+
  theme_void()
```

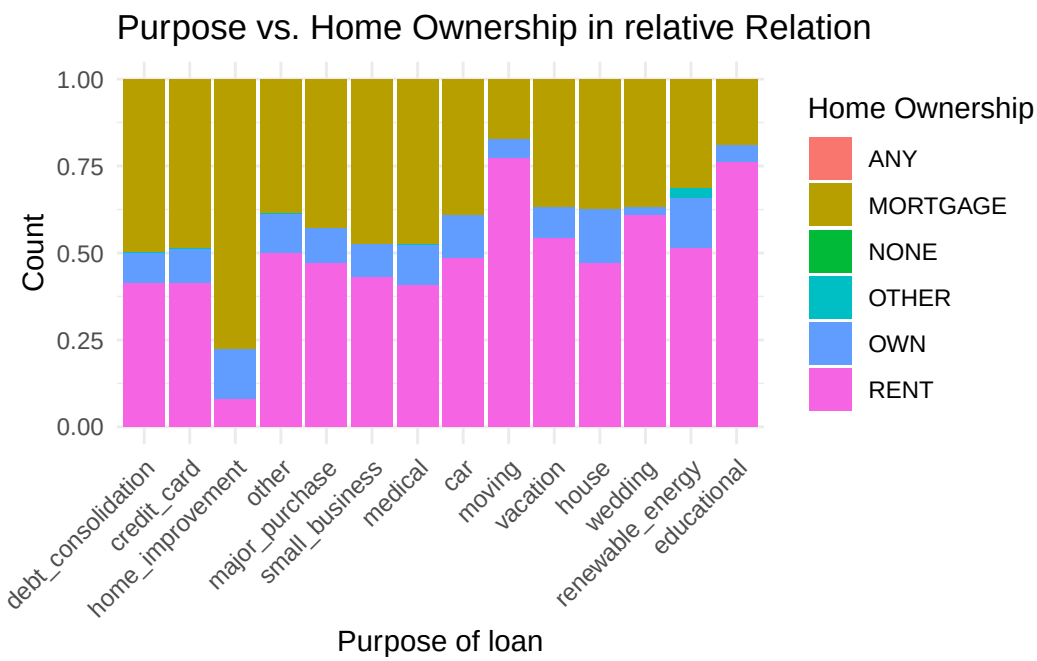
```
Ignoring unknown labels:
* file : "Home Ownership"
```




```r
ggplot(myLCdata, aes (x = forcats::fct_infreq(purpose), fill = home_ownership)) +
  geom_bar()+
  labs(
    title = "Purpose vs. Home ownership in absolute relation",
    x = "Purpose of loan",
    y = "Count",
    fill = "Home Ownership"
  )+
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```r
ggplot(myLCdata, aes (x = forcats::fct_infreq(purpose), fill = home_ownership)) +
  geom_bar( position = "fill")+
  labs(
    title = "Purpose vs. Home Ownership in relative Relation",
    x = "Purpose of loan",
    y = "Count",
    fill = "Home Ownership"
  )+
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



my thesis was wrong… People who have given as the purpose the reason "home_improvement" have as a majority a mortagege

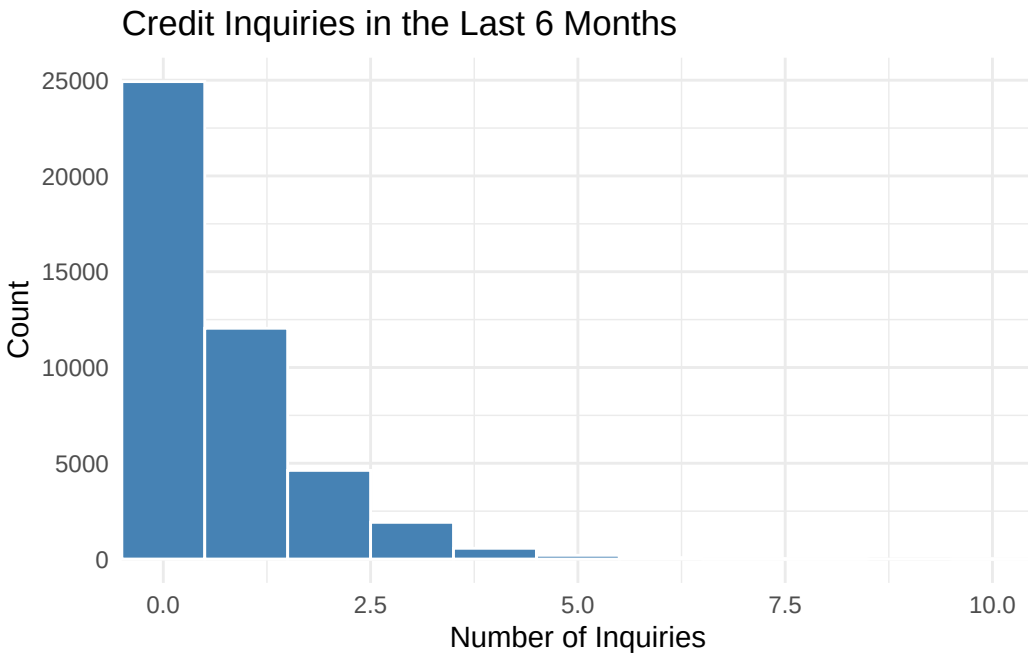
---

3.5 Variable 5: `inq_last_6mths`

3.5.1 Visualization

```
ggplot(myLCdata, aes(x = inq_last_6mths)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  coord_cartesian(xlim = c(0, 10)) +
  labs(
    title = "Credit Inquiries in the Last 6 Months",
    x = "Number of Inquiries",
    y = "Count"
  ) +
  theme_minimal()
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).
```



3.5.2 Interpretation

A histogram was chosen for this feature because it shows the frequency distribution and highlights the strong concentration of observations at low values. Here we cannot use log scale because the values are to little. The distribution of credit inquiries in the last 6 months is highly right-skewed. Most borrowers have zero or only a few inquiries, while higher numbers of inquiries occur very rarely. This results in a long tail on the right side of the distribution.

3.6 Overall Observations from Univariate EDA

- Which variables show skewness?
- Which variables show imbalance?
- Any surprising patterns?
- Do visual findings confirm summary statistics?

---

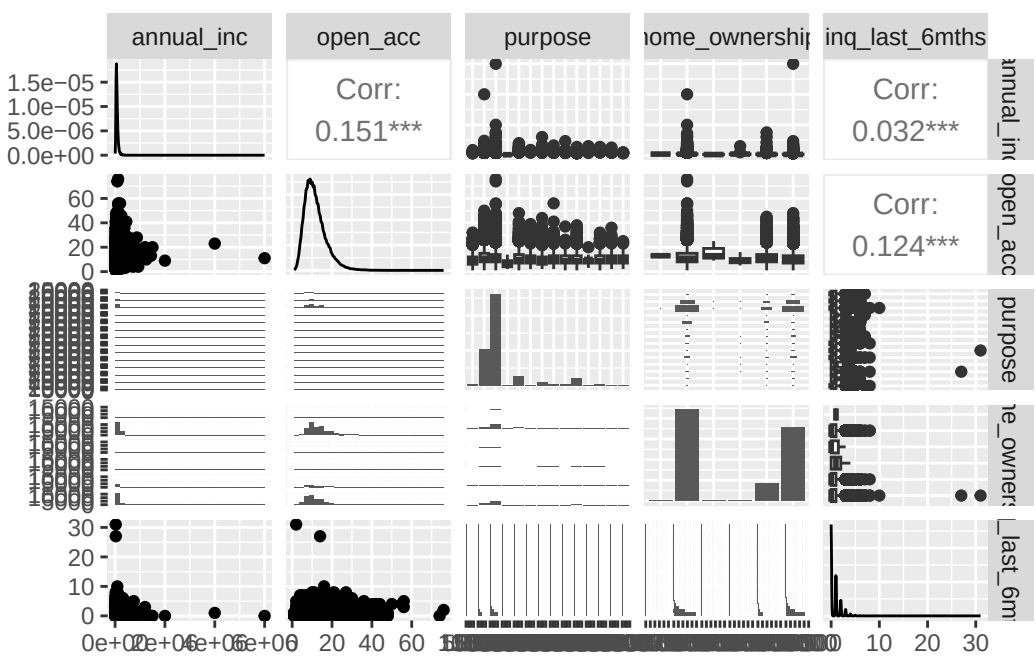
4 Task 4: Bivariate Exploratory Data Analysis (EDA)

4.1 Pairs Plot Overview

```r
if (!require("GGally")) install.packages("GGally")
library(GGally)

ggpairs(myLCdata)
```

```
Warning: Removed 2 rows containing missing values
Removed 2 rows containing missing values

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_density()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).
Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).

Warning: Removed 2 rows containing missing values

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
Warning: Removed 2 rows containing non-finite outside the scale range (`stat_bin()`).
Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_density()`).
```



4.1.1 Initial Observations

- Are there visible correlations?
- Any clear patterns?
- Any surprising relationships?

4.2 Selected Insight 1

4.2.1 Variable Pair: _____ vs _____

4.2.2 Why This Pair Is Interesting

- What pattern was observed?
- Why was it not obvious before?

4.2.3 Stand-alone Professional Plot

```
# Recreate plot using ggplot2
```

4.2.4 Interpretation

- What does this relationship indicate?
- Business interpretation?

4.3 Selected Insight 2

4.3.1 Variable Pair: _____ vs _____

4.3.2 Stand-alone Professional Plot

```
# Recreate plot using ggplot2
```

4.3.3 Interpretation

Write here.

4.4 Adding a Third Variable

4.4.1 Enhanced Plot with Third Variable (e.g., color, shape, facet)

```
# ggplot with third aesthetic mapping
```

4.4.2 Why This Third Variable?

- Why is it relevant?
- What additional insight does it provide?
- What was not visible before?

4.5 Final Bivariate Insights

- Most important relationships discovered:
- Implications for credit risk assessment:
- Potential hypotheses for predictive modeling: